M.C.A. (2nd Semester)
Cloud Computing with DevOps

Course Objective: Introduce the concepts, techniques of cloud computing and DevOps for automation, containerization, and CI/CD principles.

Laboratory Instruction:

- Students will not be permitted to attend the laboratory unless they bring the practical journal.
- Students should obtain the signature of the allocated lab teacher in the journal after completing each experiment.
- The nature of submission is handwritten for all the problem definitions.
- Students may show exercise output on his/her device.
- Prerequisite for Journal Certification: Students must show one integrated Cloud Computing and DevOps activity that covers each exercise.
- The workbook will be evaluated based on the following given rubric parameters.
    - Completeness of Solution
    - Code Correctness
    - Quality of answer
    - Coding standard
    - Demonstration
    - Technical viva
- A student must have the following computing environment in the laboratory and/or on his/her laptop.
    - GitHub
    - Docker
    - GCP authentication
    - Python library's

Practical No. 1	Enrolment No._____
Practical Problem	A. Explain cloud computing architecture with diagram and its service models and deployment models. Also, write at least four challenges of cloud computing. B. What is Google Cloud Platform (GCP)? Explain its features and provide a list of tools and services. Also, explain the steps to configure Google Cloud Platform.
Objective(s)	• Basic knowledge of cloud computing • Understanding about the Google cloud platform (GCP).
Duration for completion	**2 Hours** — **Nature of submission:** Handwritten solution
Solution must contain	Comparison appropriate detail description
References for solving the problem	NA
Sample testing data and Outcome	NA

Assessment

Evaluation Criteria	Marks	Secured by Student
Understanding of cloud computing architecture	4	
Neatness, clarity, and diagram representation	2	
Understanding of GCP Platform	2	
Technical Viva	2	
Teacher's Signature & Date		

Practical No. 2	**Enrolment No.**_____
Practical Problem	In this practical, create a simple project containing one text file and push the project to GitHub using Git commands. A. Write the GitHub/Git commands used for creating a repository, initializing Git, committing files, pushing changes, merging branches, cloning repositories, and other basic GitHub operations. B. Write the step-by-step procedure followed to complete this practical, along with the commands used and appropriate screenshots.
Objective(s)	▪ To understand the process of pushing a local project to GitHub using command-line tools. ▪ To understand the basic concepts of Git and GitHub.
Duration for completion	3 hours
Solution must contain	**Nature of submission:** Handwritten solution and attached appropriate screenshots.

Assessment

Evaluation Criteria	**Marks**	**Secured by Student**
Correct use of Git commands	3	
Creating a repository and push commands	3	
Explain the steps	2	
Technical Viva	2	
Teacher's Signature & Date		